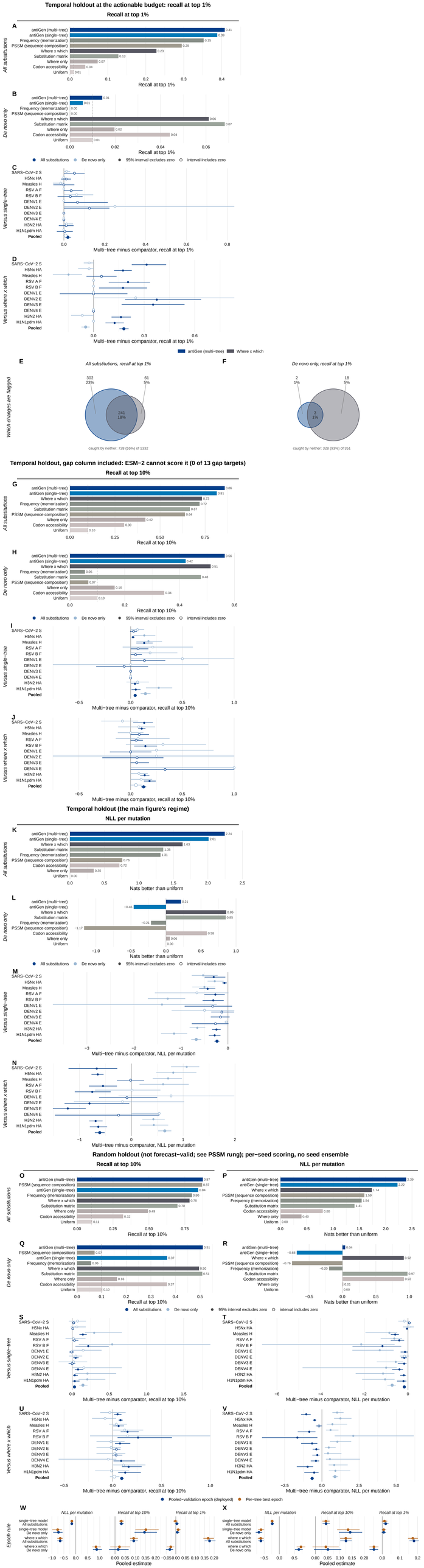
Temporal holdout at the actionable budget: recall at top 1%
Recall at top 1%
A
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
PSSM (sequence composition)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
0.41
0.39
0.35
0.29
0.23
0.13
0.07
0.04
0.01
All substitutions
Recall at top 1%
B
antiGen (multi−tree)
antiGen (single−tree)
Frequency (memorization)
PSSM (sequence composition)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
0.01
0.01
0.00
0.00
0.06
0.07
0.02
0.04
0.01
De novo only
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus single−tree
Multi−tree minus comparator, recall at top 1%
D
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus where x which
Multi−tree minus comparator, recall at top 1%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 1%
302
23%
61
5%
241
18%
caught by neither: 728 (55%) of 1332
Which changes are flagged
F
De novo only, recall at top 1%
2
1%
18
5%
3
1%
caught by neither: 328 (93%) of 351
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
G
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.86
0.81
0.73
0.72
0.67
0.64
0.42
0.30
0.10
All substitutions
Recall at top 10%
H
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.56
0.42
0.51
0.05
0.48
0.07
0.16
0.34
0.10
De novo only
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
I
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus single−tree
Multi−tree minus comparator, recall at top 10%
J
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus where x which
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
K
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Substitution matrix
Frequency (memorization)
PSSM (sequence composition)
Codon accessibility
Where only
Uniform
2.24
2.01
1.63
1.35
1.31
0.76
0.72
0.35
0.00
All substitutions
Nats better than uniform
L
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Substitution matrix
Frequency (memorization)
PSSM (sequence composition)
Codon accessibility
Where only
Uniform
0.21
−0.46
0.86
0.85
−0.21
−1.17
0.58
0.06
0.00
De novo only
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
M
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus single−tree
Multi−tree minus comparator, NLL per mutation
N
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus where x which
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
O
antiGen (multi−tree)
PSSM (sequence composition)
antiGen (single−tree)
Frequency (memorization)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
0.87
0.87
0.84
0.80
0.76
0.70
0.49
0.32
0.11
All substitutions
Recall at top 10%
NLL per mutation
P
antiGen (multi−tree)
antiGen (single−tree)
Where x which
PSSM (sequence composition)
Frequency (memorization)
Substitution matrix
Codon accessibility
Where only
Uniform
2.39
2.22
1.74
1.59
1.54
1.41
0.80
0.40
0.00
Nats better than uniform
Q
antiGen (multi−tree)
PSSM (sequence composition)
antiGen (single−tree)
Frequency (memorization)
Where x which
Substitution matrix
Where only
Codon accessibility
Uniform
0.51
0.07
0.37
0.06
0.50
0.51
0.16
0.37
0.10
De novo only
Recall at top 10%
R
antiGen (multi−tree)
antiGen (single−tree)
Where x which
PSSM (sequence composition)
Frequency (memorization)
Substitution matrix
Codon accessibility
Where only
Uniform
0.04
−0.88
0.92
−0.76
−0.20
0.97
0.92
0.03
0.00
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
S
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus single−tree
Multi−tree minus comparator, recall at top 10%
T
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Multi−tree minus comparator, NLL per mutation
U
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Versus where x which
Multi−tree minus comparator, recall at top 10%
V
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
Pooled
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
W
NLL per mutation
Recall at top 10%
Recall at top 1%
Epoch rule
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
Pooled estimate
X
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
Pooled estimate